



FRANKFURT UNIVERSITY OF APPLIED SCIENCES
FACULTY OF ENGINEERING – COMPUTER SCIENCE PROGRAM

Traffic Simulation - Java Project

JAVAFX-BASED SUMO TRAFFIC SIMULATION CONTROLLER

Group Number: Tuesday Exercise
(8:15-9:45am) - Group 4

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2. Lars Michael Keppler
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Course: Object-Oriented Programming
in Java

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Abstract

This project presents a JavaFX-based graphical user interface for controlling and visualizing traffic simulations in SUMO (Simulation of Urban Mobility). The application allows users to inject vehicles, controlling traffic lights, and interact with the simulation in real time. The system is modular, extensible, and leverages the Libtraci library for SUMO connectivity.

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Declaration of Authorship

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Lars Michael Keppler Frankfurt am Main, 23.11.2025

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Lars Michael Keppler

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Abel Solyom Duisburg, 24.11.25

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Minh Luan Vo Frankfurt am Main, 23.11.2025

First and last name City, date and signature

Minh Luan Vo

We have used AI tools (i.e. ChatGPT) for research purposes. That is, we have used them to get ideas about libraries such as JavaFX or understanding errors.

1 Introduction

1.1 Project Overview

SUMO_Group4 is a Java application that provides a user-friendly interface for managing and visualizing traffic simulations using the SUMO platform. The application is built with JavaFX for the GUI and uses the Libtraci library to communicate with SUMO via the TraCI protocol.

2 Project Structure

2.1 Directory Layout

- **pom.xml**: Maven configuration file.
- **lib/libtraci.jar**: External library for SUMO connectivity.
- **src/main/java/com/example/gui**: Java source code.
- **src/main/resources/com/example/gui**: FXML UI, images, SUMO config files.
- **target/**: Compiled output.
- **README.md**, **projectDocumentation.pdf**: Documentation.

2.2 Key Classes and Files

- **Launcher.java**: Application launcher.
- **Main.java**: Application initializer.
- **GUI.java**: JavaFX controller.
- **SimulationController.java**: Manages SUMO connection.
- **VehicleController.java**: Controls Vehicles in the Simulation.
- **TrafficLightController.java**: Controls TrafficLights in the Simulation.
- **VehicleModel.java**: Model class for Vehicles.
- **TrafficLightModel.java**: Model class for Traffic Lights.
- **LaneLayer.java**: Manages the rendering of Lanes.
- **CarLayer.java**: Manages the rendering of Vehicles.
- **TrafficLightLayer.java**: Manages the rendering of Traffic Lights.
- **MapUtil.java**: Scales the rendering inside the GUI.
- **SumoConnection.java**: TraCI protocol helper.
- **Statistics.java**: Manages statistics of the Simulation.
- **MapSumoConfig.java**: Handles SUMO config files.
- **GUI-view.fxml**: GUI layout.
- SUMO config files: **.net.xml**, **.rou.xml**, **.sumocfg**.

3 Implementation Details

3.1 Main Application Flow

```
public void start(Stage stage) throws Exception {
    FXMLLoader fxmlLoader = new FXMLLoader(Main.class.getResource("
        GUI-view.fxml"));
    Scene scene = new Scene(fxmlLoader.load());
    stage.setTitle("SumoController");
    stage.setScene(scene);
    stage.show();

    GUI gui = fxmlLoader.getController();

    SimulationController simController = new SimulationController();
    gui.setSimulationController(simController);
    simController.makeConnection();

    Statistics stat = new Statistics(simController);
}
```

3.2 GUI Example

```
@FXML
public void commandSpawnVehicle(ActionEvent e) {
    System.out.println("Spawning new vehicle!");
    simController.spawnVehicle();
}
```

4 Class Relationships

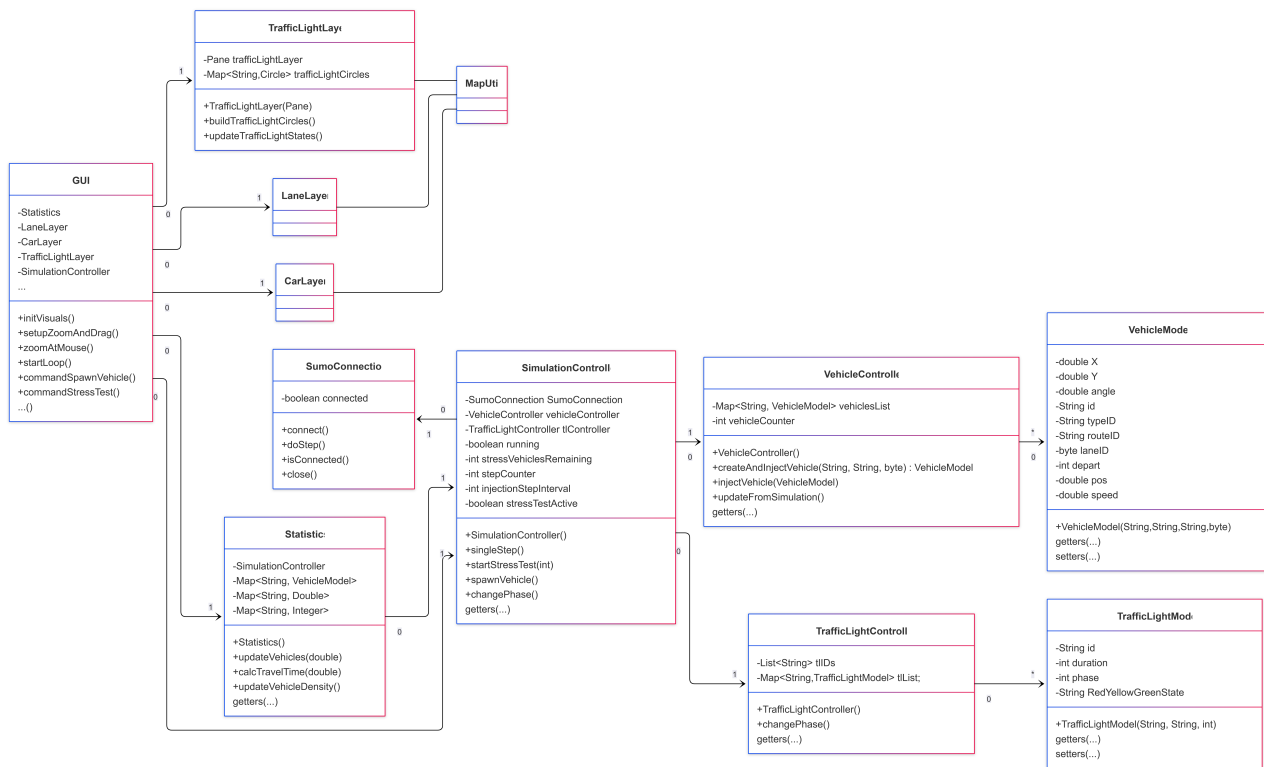


Figure 1: UML Class Diagram

5 Results and Discussion

5.1 Features

- Simulation visualization in our application.
- Real-time vehicle injection in SUMO.
- Real-time traffic light controlling.
- Stress-test function to trigger injection of multiple vehicles at once.
- Statistics section in the GUI to track the average speed of all vehicles in the Simulation.
- Modular and extensible architecture.
- JavaFX + FXML for maintainability.

5.2 Current Limitations and Ongoing Development

- Providing more statistics in the statistic section.
- Vehicle grouping/filtering.
- Exportable reports for statistics.
- Usage of more Threads.
- Support of more vehicle types.

6 Conclusion

Our project provides a robust and user-friendly interface for SUMO traffic simulation. The modular design and Java technologies make it suitable for further development.